

Digital Design

FSM Optimization

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Outline

- FSM Optimization: **State Minimization**
 - **Row Matching method**
 - **Partitioning method**
 - Implication Chart
- FSM Optimization : State Encoding

FSM Optimization

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- FSM Optimization: **State Minimization**
 - **Row Matching method**
 - **Partitioning method**
 - Implication Chart
- FSM Optimization : State Encoding

FSM : State Minimization

- Minimizing number of state reduce
 - Requirement of bigger size state register
 - Possibly reduce the Combinational Circuit Complexity (CCC) or **NS logic** for the FSM

References materials

- **Section 14.6 and 14.7 of Kumar Book**
 - A. Anand Kumar, *Fundamentals of Digital Circuits* 3rd Edition, PHI. 2014 **(This book have a lot of examples to understand the concepts)**
- **Section 8.1 of Katz Book**
 - Randy H. Katz, G Borriello, *Contemporary Logic Design*, 2nd Edition, PHI, India, 2009
 - This is one of the prescribed/listed text book for the course along with Mano Book

Some Definitions

- **State Equivalence:** S1 and S2 are equivalent if for every input sequence applied to machine goes to same NS and Output
 - If $S1(t+1)=S2(t+1)$ and $Z1=Z2$ then $S1=S2$
- **Distinguishable States:** Two states S1 and S2 are Distinguishable *if and only if* there exist at least one finite input sequence which produce different outputs from S1 and S2

FSM Optimization Methods

- FSM State Optimization
 - Given FSM
 - Goal is to reduce number of states of FSM
- Method 1: Row Matching Method
 - Completely specified machine (n^2 edges)
 - Partially specified machine
- Method 2: Partitioning Method
- Method 3: Implication Chart Method

Row Matching Methods

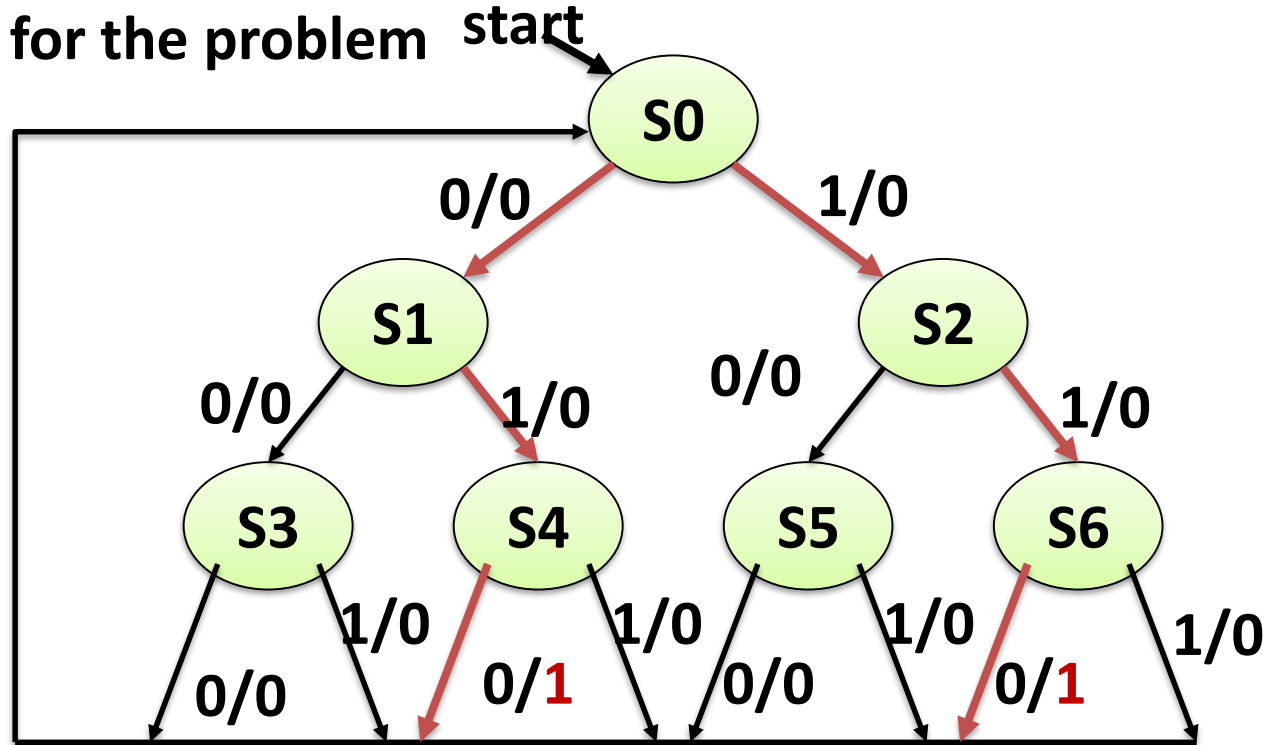
State Minimization : Example

- **Sequence Detector for 010 or 110 :**
 - After each 3 bit input sequence
 - if it consist of 110 or 010, out put 1

State Minimization : Example

- Sequence Detector for 010 or 110:
 - After each 3 bit input sequence , if it consist of 110 or 010, out put 1
- Your designed FSM for the problem start

Is it
optimized?



State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S4 S5 S6)

State Minimization Example

- S4 and S6 are different as CMP to other States based on output

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S5) (S4 S6)

State Minimization Example

- S4 and S6 have same NS and O/P, they are same

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S5) (S4 S6)

State Minimization Example

- S4 and S6 have same NS and O/P, they are same

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4'	0	0
1	S2	S5	S4'	0	0
00	S3	S0	S0	0	0
01	S4'	S0	S0	1	0
10	S5	S0	S0	0	0
11	S4'	S0	S0	1	0

(S0 S1 S2 S3 S5) (S4')

State Minimization Example

- Reduced FSM after merging S4, S6 to S4'

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4'	0	0
1	S2	S5	S4'	0	0
00	S3	S0	S0	0	0
01/11	S4'	S0	S0	1	0
10	S5	S0	S0	0	0

(S0 S1 S2 S3 S5) (S4')

State Minimization Example

- NS of S1, S2 are different as compared to S0, S3 and S5

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4'	0	0
1	S2	S5	S4'	0	0
00	S3	S0	S0	0	0
01/11	S4'	S0	S0	1	0
10	S5	S0	S0	0	0

(S0 S1 S2 S3 S5) (**S4'**)

State Minimization Example

- NS of S1, S2 are different as compared to S0, S3 and S5

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4'	0	0
1	S2	S5	S4'	0	0
00	S3	S0	S0	0	0
01/11	S4'	S0	S0	1	0
10	S5	S0	S0	0	0

(S1 S2) (S0 S3 S5) (S4')

State Minimization Example

- S3 and S5 have same NS and O/p, so they are same

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4'	0	0
1	S2	S5	S4'	0	0
00	S3	S0	S0	0	0
01/11	S4'	S0	S0	1	0
10	S5	S0	S0	0	0

(S1 S2) (S0 S3 S5) (S4')

State Minimization Example

- Merging S3 and S5 to **S3'**

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3'	S4'	0	0
1	S2	S3'	S4'	0	0
00/10	S3'	S0	S0	0	0
01/11	S4'	S0	S0	1	0

(S1 S2) (S0 S3') (S4')

State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3'	S4'	0	0
1	S2	S3'	S4'	0	0
00/10	S3'	S0	S0	0	0
01/11	S4'	S0	S0	1	0

(S1 S2) (S0 S3') (S4')

State Minimization Example

- S1 and S2 have same NS and O/P, they are same

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3'	S4'	0	0
1	S2	S3'	S4'	0	0
00/10	S3'	S0	S0	0	0
01/11	S4'	S0	S0	1	0

(S1 S2) (S0 S3') (S4')

State Minimization Example

- Merging S1 and S2 to S1'

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1'	S1'	0	0
0/1	S1'	S3'	S4'	0	0
00/10	S3'	S0	S0	0	0
01/11	S4'	S0	S0	1	0

(S1') (S0 S3') (S4')

State Minimization Example

- S_0, S_3' are different state as their NS are different

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S_0	S_1'	S_1'	0	0
0/1	S_1'	S_3'	S_4'	0	0
00/10	S_3'	S_0	S_0	0	0
01/11	S_4'	S_0	S_0	1	0

(S_1') (S_0 S_3') (S_4')

State Minimization Example

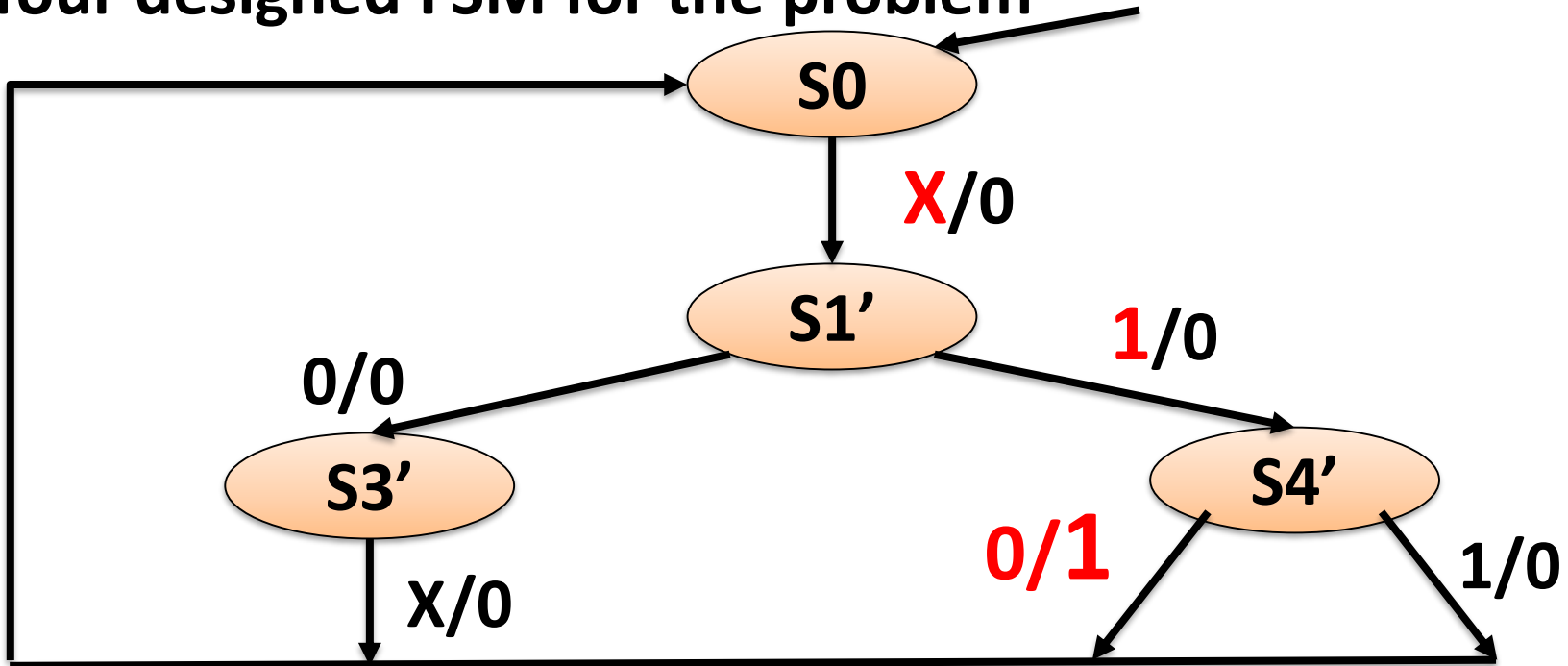
- No further matching $\Rightarrow$ Reduced one

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1'	S1'	0	0
0/1	S1'	S3'	S4'	0	0
00/10	S3'	S0	S0	0	0
01/11	S4'	S0	S0	1	0

(S1') (S0) (S3') (S4')

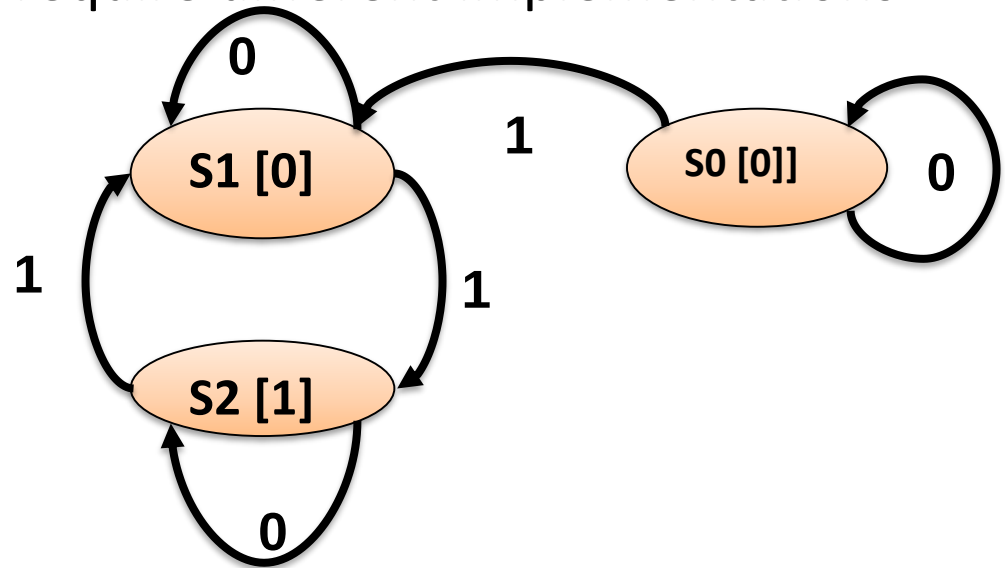
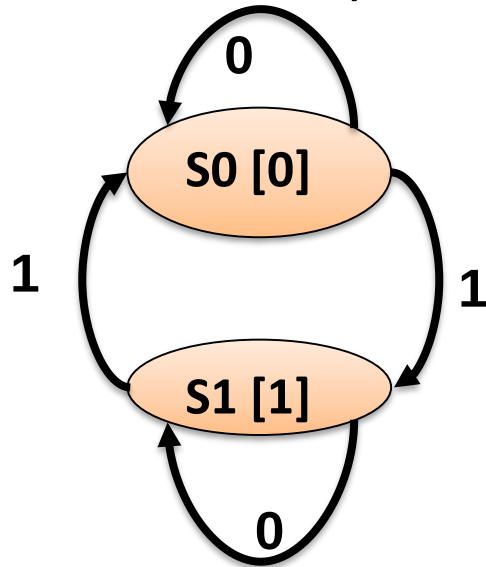
Minimized FSM

- Sequence Detector for 010 or 110: After each 3 bit input sequence, if it consists of 110 or 010, output 1
- Your designed FSM for the problem



Row Matching Method : Fallacies

- Odd Parity Checker:
 - Two alternative state diagrams
 - Identical output behavior on all input strings
 - FSMs are *equivalent*, but require different implementations



Critique of Row Matching

- Straightforward to understand and easy to implement
- **Problem:** does not allow yield the most reduced state table!

Present State	Next State		Output	(S0 S1 S2) Based on output (S0 S2) (S1)
	X=0	X=1		
S ₀	S ₀	S ₁	0	
S ₁	S ₁	S ₂	1	
S ₂	S ₂	S ₁	0	

No way to combine states S0 and S2
based on Next State Criterion!

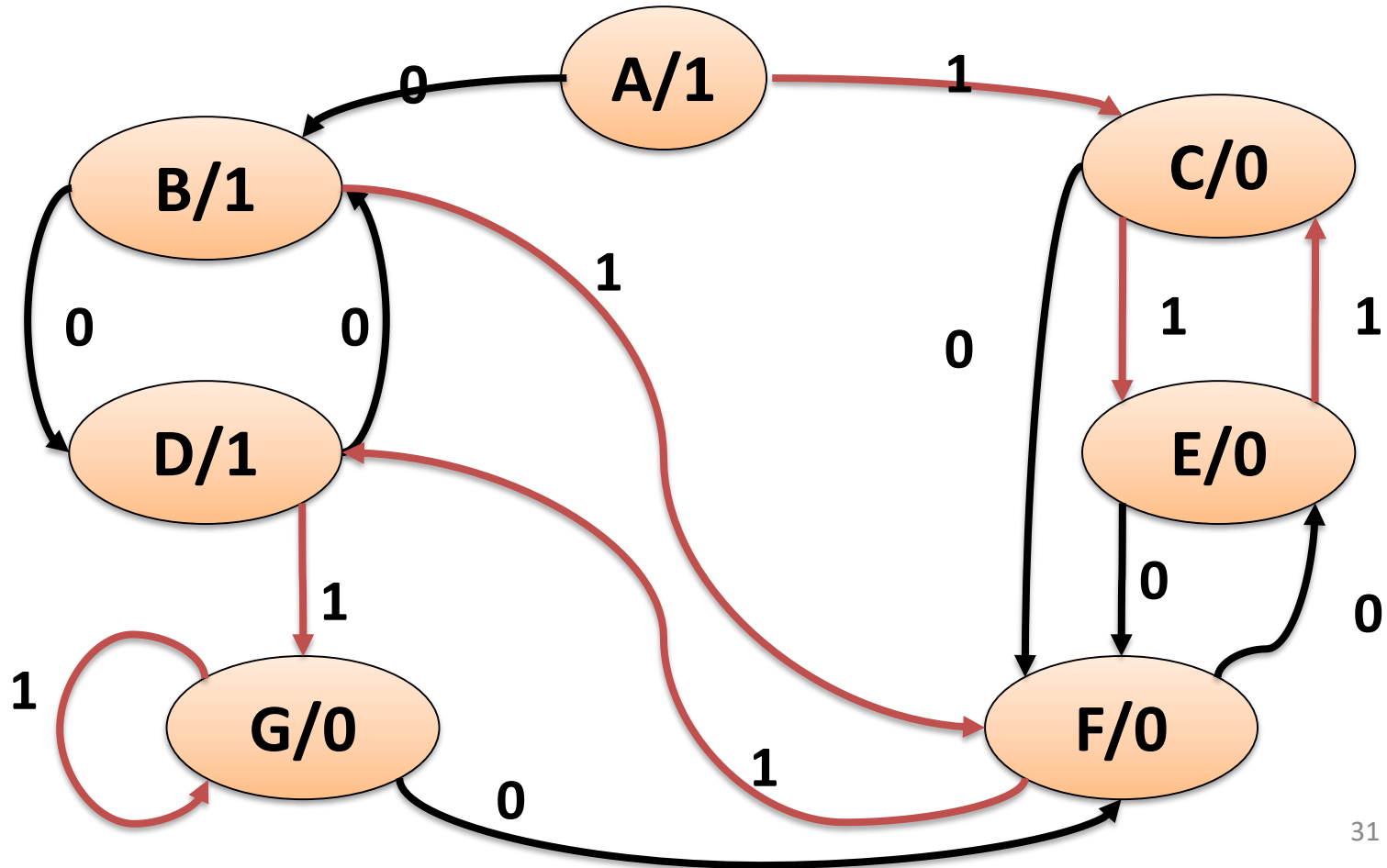
Partitioning Methods

State Minimization: Partitioning

- Form an initial partition (P_1) : that includes **all the** states.
- Form a 2nd partition (P_2)
 - By separating the states into two blocks based upon their output values.
- Form a third partition (P_3)
 - by separating the states into blocks corresponding to the next state values.
- Continue partitioning until
 - Two successive partitions are the same (i.e. $P_{N-1} = P_N$).
- All states in any one block are equivalent
 - Equivalent states can be combined into a single state.

State Minimization Partitioning: Example

State
Diagram



State Minimization: Partitioning

Present State	Next State		Output Z
	W=0	W=1	
A	B	C	1
B	D	F	1
C	F	E	0
D	B	G	1
E	F	C	0
F	E	D	0
G	F	G	0

State Minimization: Partitioning

Initial Partition:

$$P_1 = (ABCDEFGG)$$

The initial partition contains all states
in the state diagram / table.

State Minimization: Partitioning

- Separate states based on output value.

$$P_2 = (\text{ABD})(\text{CEFG})$$

Present State	Next State		Output Z
	W=0	W=1	
A	B	C	1
B	D	F	1
C	F	E	0
D	B	G	1
E	F	C	0
F	E	D	0
G	F	G	0

State Minimization: Partitioning

$$P_2 = (\text{ABD}) \ (\text{CEFG})$$

- Separate states based on next state values.



$$P_3 = (\text{ABD}) \ (\text{CEG}) \ (\text{F})$$

unique state

State Minimization: Partitioning

$$P_3 = (ABD) (CEG) (F)$$

- Separate states based on next state values.



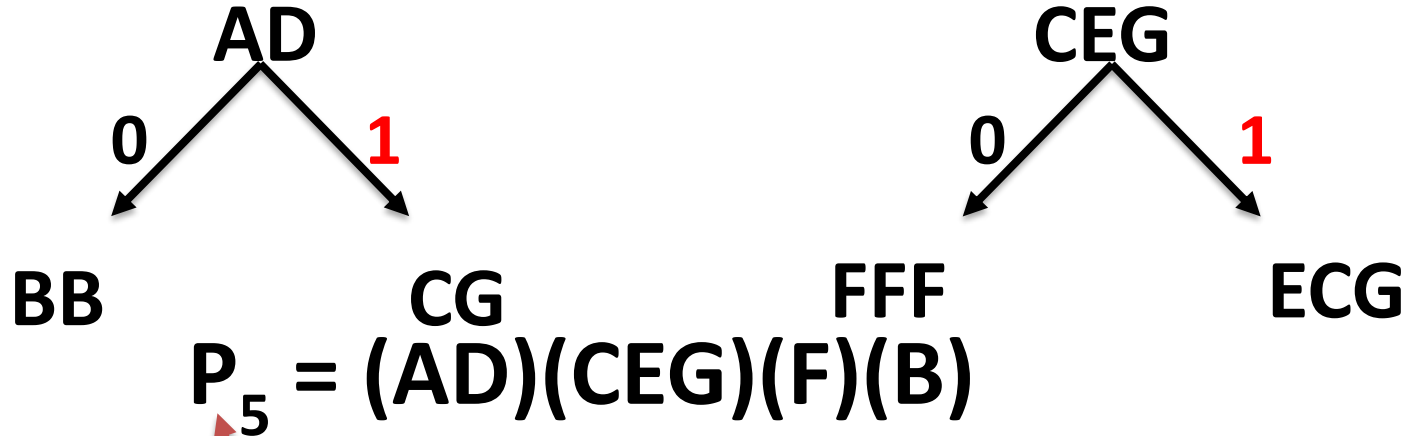
$$P_4 = (AD) (CEG) (F) (\textcolor{red}{B})$$

unique state

State Minimization: Partitioning

$$P_4 = (AD) (CEG) (F) (B)$$

- Separate states based on next state values.



Same as previous partition (P_4)

State Minimization: Partitioning

- Since $P_4 = P_5$, state minimization is complete.
- The equivalent states are:
 - **A = D**
 - **C = E = G**
 - **B**
 - **F**
- Thus, the FSM can be realized with just 4 states.

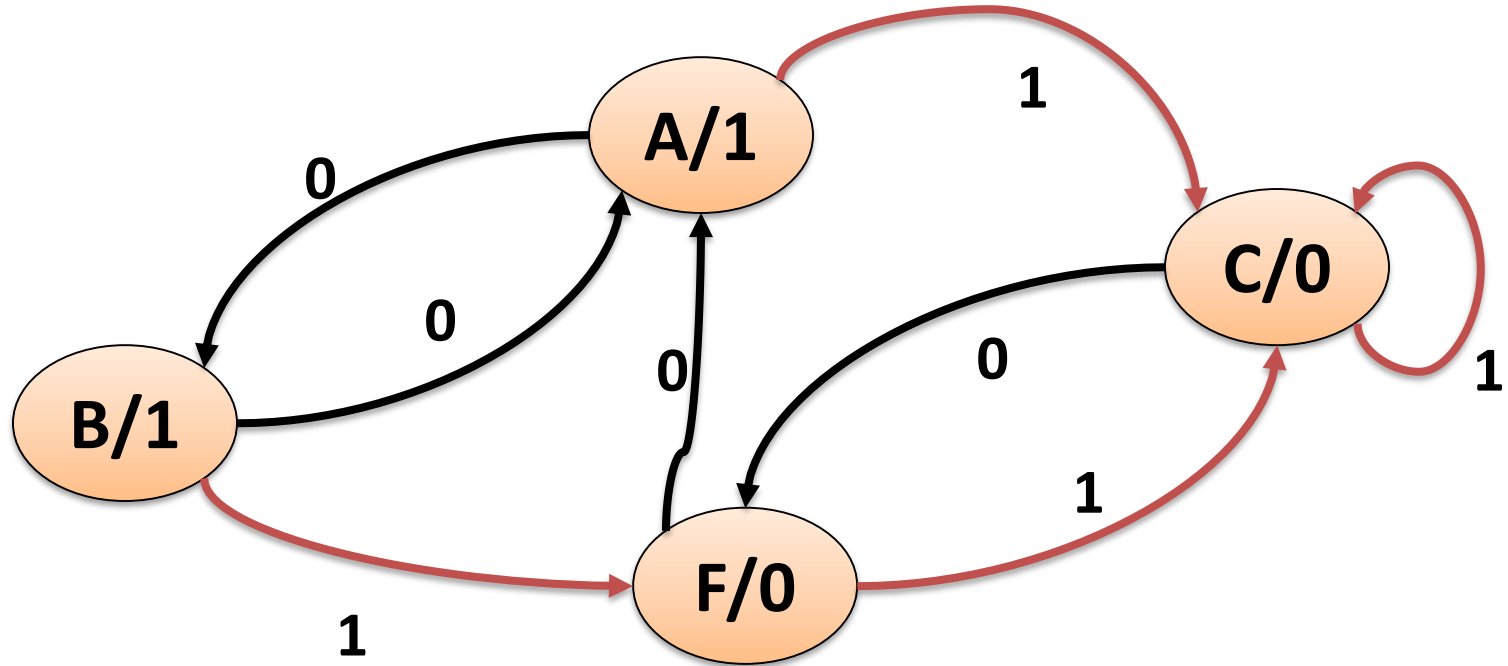
$$P_4 = (AD) (CEG) (F) (B)$$

FSM: State Minimization

Present State	Next State		Output Z
	W=0	W=1	
A	B	C	1
B	A	F	1
C	F	C	0
F	C	A	0

Minimized State Table

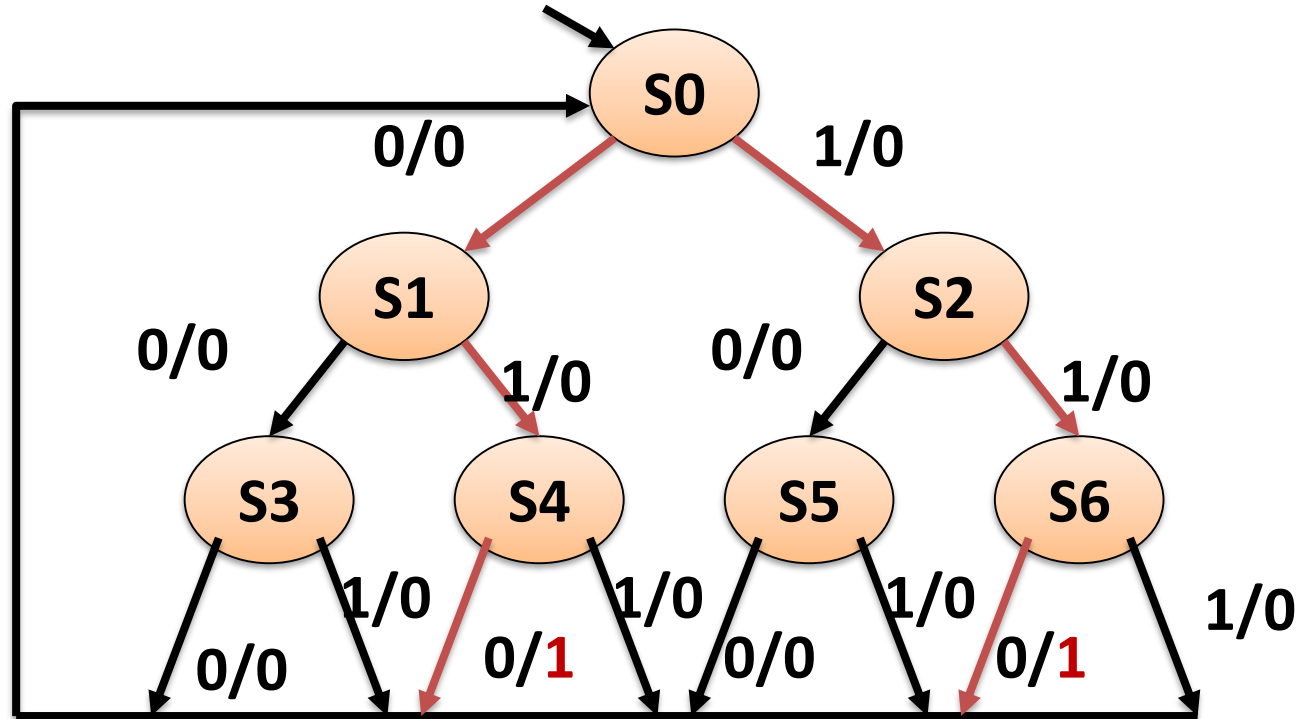
FSM: State Minimization



Minimized State Diagram

State Minimization: Previous Example

- Sequence Detector for 010 or 110
- After is asserted after each 3 bit input sequence if it consist of 110 or 010



State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S4 S5 S6)

State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S4 S5 S6)

(S0 S1 S2 S3 S5) (S4 S6)

State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S5)

0

1

(S1 S3 S5 S0 S0)

(S2 S4 S6

S0 S0)

(S0 S1 S2 S3 S4 S5 S6)

(S0 S1 S2 S3 S5)

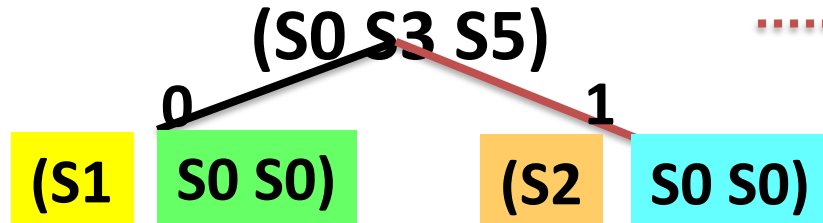
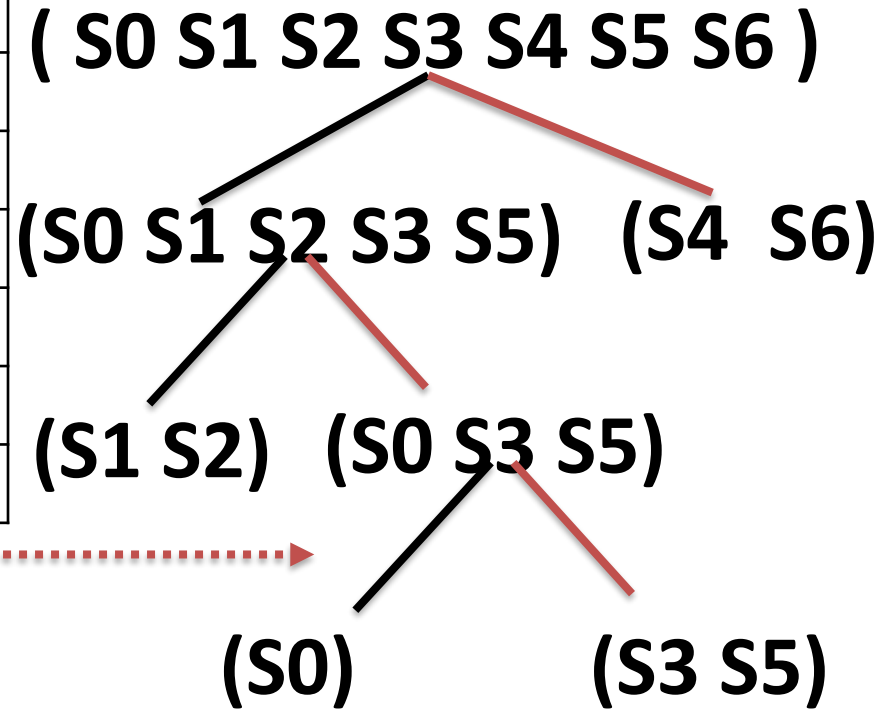
(S4 S6)

(S1 S2)

(S0 S3 S5)

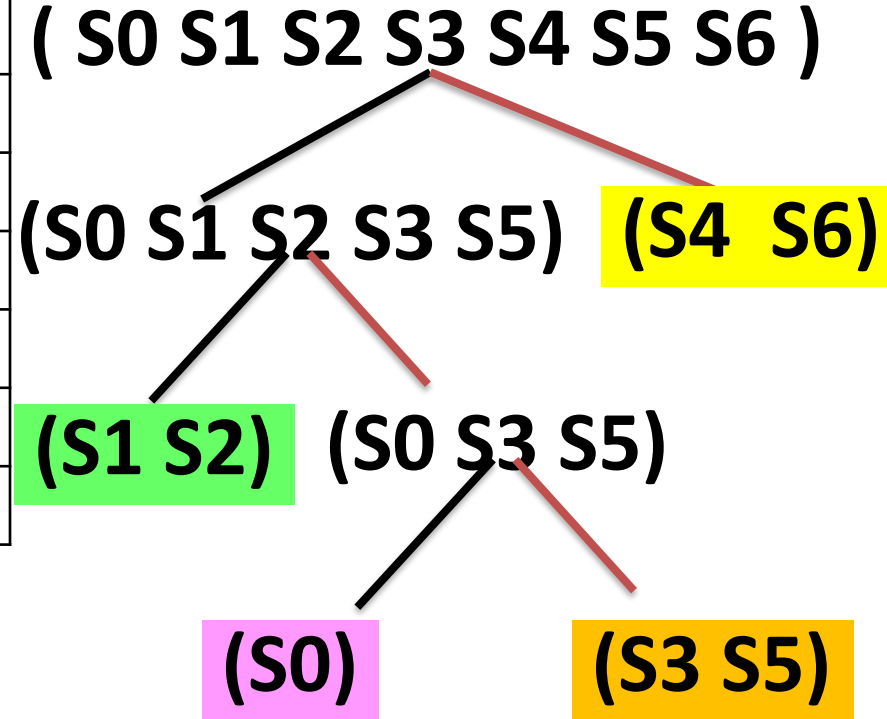
State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0



State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0



State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S4 S5 S6)

(S0)

(S4 S6)

0

1

(S0)

(S0)

(S1 S2)

0

1

(S3 S5)

(S4 S6)

(S3 S5)

0

1

(S0)

(S0)

State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

No further
partitions possible

(S0 S1 S2 S3 S4 S5 S6)

(S0)

(S4 S6)

0

1

(S0)

(S0)

(S1 S2)

0

1

(S3 S5)

(S4 S6)

(S3 S5)

0

1

(S0)

(S0)

State Minimization Example

Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1	S2	0	0
0	S1	S3	S4	0	0
1	S2	S5	S6	0	0
00	S3	S0	S0	0	0
01	S4	S0	S0	1	0
10	S5	S0	S0	0	0
11	S6	S0	S0	1	0

(S0 S1 S2 S3 S4 S5 S6)



(S0) (S1 S2) (S3 S5) (S4 S6)



(S0) (S1') (S3') (S4')

State Minimization Example

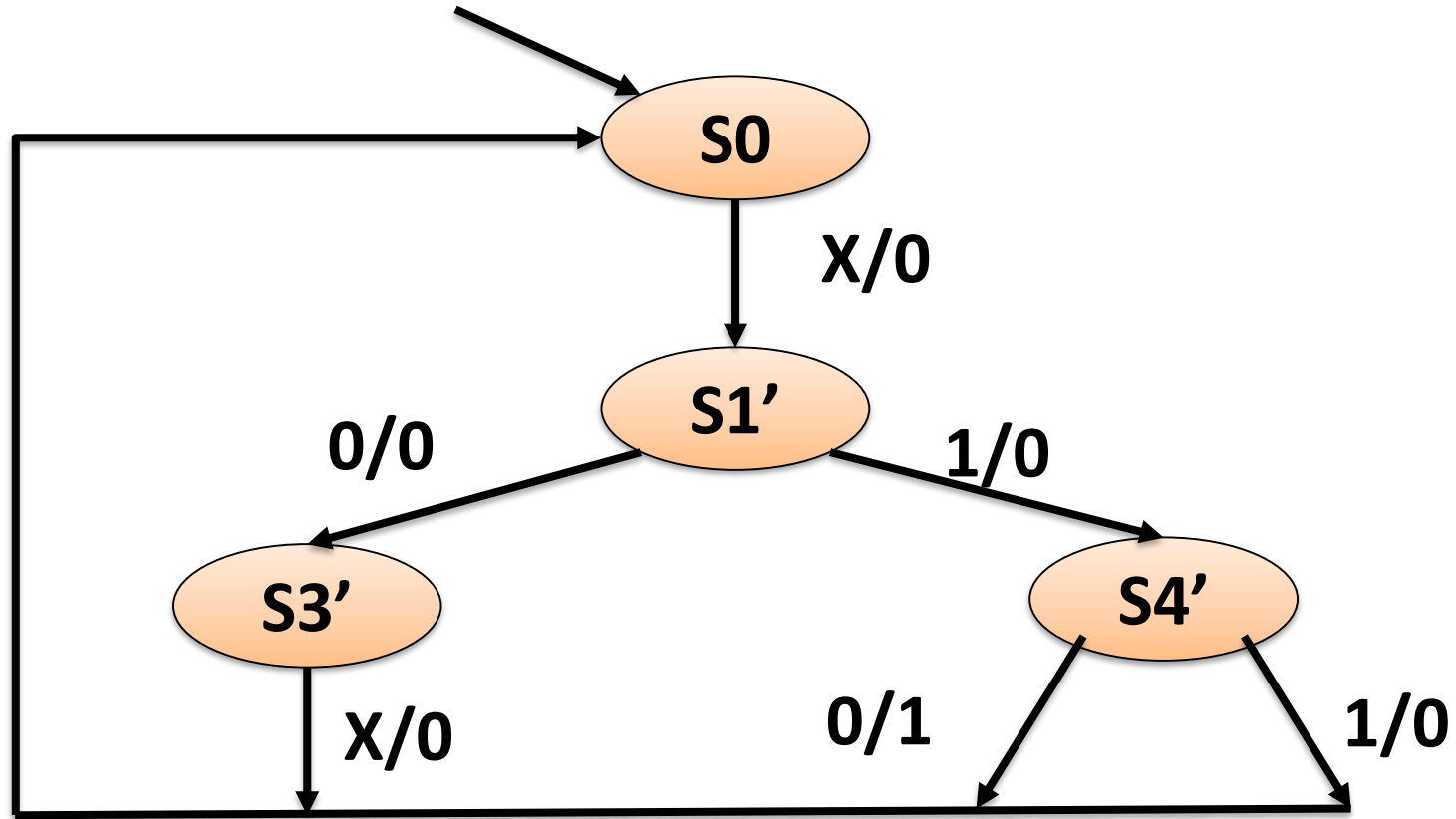
Input	PS	NS		OUTPUT	
		X=0	X=1	X=0	X=1
Reset	S0	S1'	S1'	0	0
0	S1'	S3	S4'	0	0
1	S2	S5	S4'	0	0
00	S3	S0	S0	0	0
01	S4'	S0	S0	1	0
10	S5	S0	S0	0	0
11	S4'	S0	S0	1	0

(S0) (S1 S2) (S3 S5) (S4 S6)



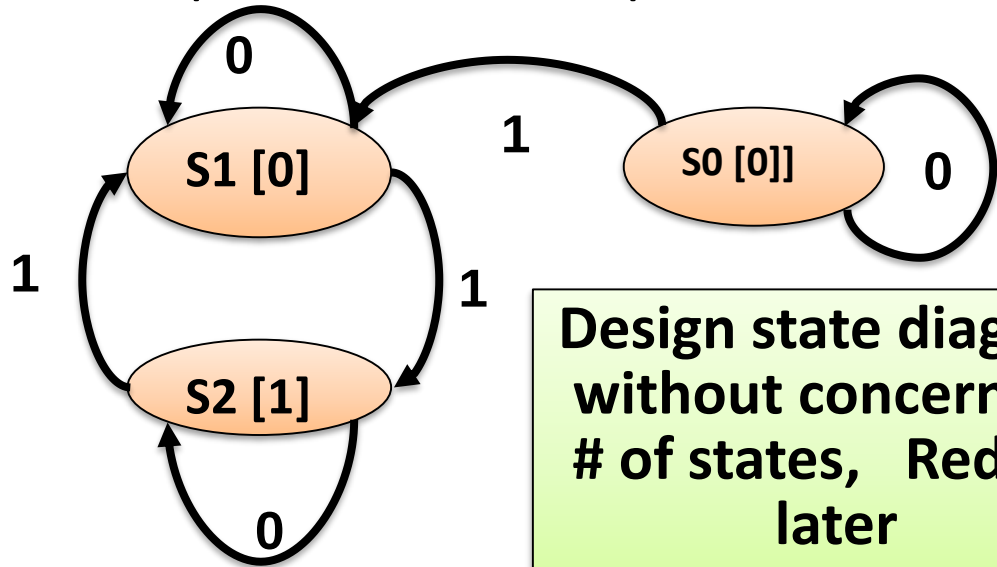
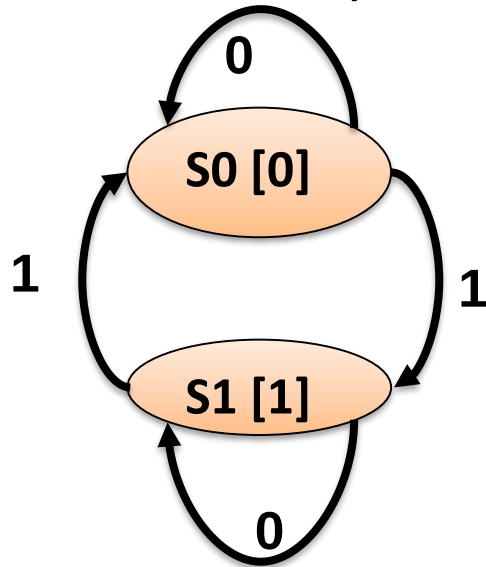
(S0) (S1') (S3') (S4')

Minimized FSM



Row Matching Method : Fallacies

- Odd Parity Checker:
 - Two alternative state diagrams
 - Identical output behavior on all input strings
 - FSMs are *equivalent*, but require different implementations



**Design state diagram
without concern for
of states, Reduce
later**

Partitioning Method

Present State	Next State		Output
	X=0	X=1	
s_0	s_0	s_1	0
s_1	s_1	s_2	1
s_2	s_2	s_1	0

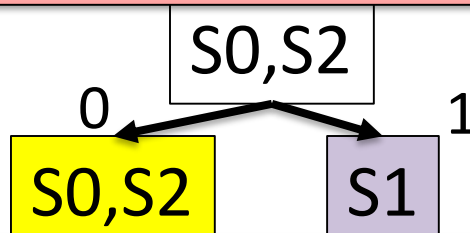
($s_0s_1s_2$)
Based on output
(s_0s_2) (s_1)

Partitioning Method

Present State	Next State		Output
	X=0	X=1	
s_0	s_0	s_1	0
s_1	s_1	s_2	1
s_2	s_2	s_1	0

($s_0s_1s_2$)
Based on output
(s_0s_2) (s_1)

Was not possible with Row matching method



s_0, s_2 are same

$\{s_0 s_2\}, \{s_1\} \Rightarrow \{s_0'\}, \{s_1\}$

But possible with Partitioning method

Thanks